

Code-Layout, Readability and Reusability

Date	05 July 2026
Team ID	XXXXXX
Project Name	Voice-Based Concept Understanding Analyser (VBCUA)
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Yes	Python formatting standards followed
2	Proper File Structure	Files and folders are logically organized	Yes	Separate folders for modules, assets, reports and documentation
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	Clear names used for audio, scores, and analysis data
4	Function / Method Names	Functions are descriptively named	Yes	Functions represent specific processing tasks
5	Code Comments	Inline and block comments explain logic	Yes	Important AI processing steps documented
6	Modular Design	Code is split into reusable functions/modules	Yes	Speech, analysis, UI and report modules separated
7	No Redundant Code	Duplicate or unused code is removed	Yes	Code optimized during testing phase
8	Error Handling	Exceptions and errors are handled gracefully	Yes	Audio upload and processing errors handled

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
1	Speech-to-Text Module	Python, OpenAI Whisper	Voice processing section	High

2	Semantic Analysis Module	Python, Sentence Transformers	Concept evaluation engine	High
3	Audio Feature Module	Python, Librosa	Speech quality analysis	High
4	Report Generation Module	Python, ReportLab	PDF feedback reports	Medium
5	User Interface Components	Streamlit	Application pages and dashboard	Medium

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Well-organized modular project structure
Readability	5	Clear naming conventions and understandable code
Reusability	5	Reusable AI processing modules implemented
Documentation / Comments	4	Important sections documented properly
Overall Score	5	Application follows good development practices